HRM Pro Enterprise

SYSTEM ARCHITECTURE & COMPREHENSIVE PRODUCT ANALYSIS REPORT

Document Class: Architecture & Product Specification
Author: Senior Software Architect & Product Analyst
Version: 4.2.0 • Release Date: October 2026
Environment: Hybrid Cloud (PostgreSQL / Express / Vanilla SPA)

Executive Architectural Summary: HRM Pro is an enterprise Human Resource Management System engineered specifically for multi-subsidiary corporate holdings and Pakistan statutory compliance (FBR 2025–27 Progressive Tax Slabs, 30/26 Gratuity exit settlements, 6 standard bank disbursal formats). The system operates as an offline-first Single Page Application (SPA) with dual-mode data synchronization (local in-memory/localStorage cache + central Express.js REST API, real-time WebSockets, and Supabase/Prisma PostgreSQL).

1. Application Overview

1.1 Application Name & Identity

HRM Pro (MY-HRM Enterprise) — Cloud Workforce Intelligence & Statutory Operations Management Suite.

1.2 Business Goal & Purpose

The primary business objective is to eliminate error-prone spreadsheet payroll calculations, streamline attendance fraud, enforce Pakistan statutory labor laws, and unify multi-company holding operations under a single corporate console with strict role isolation.

1.3 Target Users

- **Super Administrators & Group CXOs:** Holding-wide visibility, NTN profiles, entity switcher, global audit trails.
- **HR Directors & Managers:** Employee master files, e-DMS vaults, recruitment ATS, leave authorization, 4-gate exit clearances.
- **Finance & Payroll Accountants:** FBR 2025–27 progressive income tax brackets, EOBI/PF cumulative ledgers, 1Link/PayPak bank advice.
- **Department Line Managers:** Department rosters, attendance regularization, peer shift swaps, team performance reviews.
- **Employees & New Recruits:** Self-service punch clock, payslip downloads, leave requests, expense lodging, induction checklists.

1.4 Main Problems Solved

Problem Domain	Legacy / Manual Challenge	HRM Pro Architectural Solution
Pakistan Tax Compliance	Complex progressive slabs (0%–35%) calculated in Excel leading to audit disputes with the FBR.	Dynamic FBR 2025–27 Tax Engine with mid-year increment adjustments, marginal relief, and tax certificates.
Statutory Gratuity Offboarding	Subjective calculations of final dues and uncoordinated equipment returns causing legal risk.	Automated `(Last Gross ÷ 26 × 30) × Years` engine with 4-gate digital sign-offs (HR, IT, Finance, Admin).
Attendance & Ghost Punches	Manual paper muster rolls and disconnected biometric USB thumb drives.	Native TCP/IP hardware push (ZKTeco/SilkID), terminal simulator, offline PWA queue, and GPS geofence radar.
Multi-Company Holding Chaos	Multiple legal entities sharing unisolated spreadsheets risking confidentiality.	Model A Holding Scoping: Subsidiary HR managers locked to their legal entity; Super Admins get cross-NTN global telemetry.
Internal Communications	Scattered WhatsApp groups and lost email threads for operational updates.	Integrated Microsoft Teams-style Collaboration Hub with department channels, 1-on-1 direct messaging, and video meeting room launcher.

1.5 Overall System Workflow

The end-to-end operational lifecycle progresses through six sequential phases:

1. **Talent Acquisition:** Job requisition → Public careers application → 5-Stage ATS Kanban → 10-Criteria rubric scoring → Automated PDF offer letters.
2. **Digital Onboarding & e-DMS:** Induction vault → CNIC, blood group, degrees upload → 30/60/90-day expiry radar → Contract digital acknowledgment.

3. **Time & Attendance Tracking:** Biometric hardware punch / Geotagged web clock → Grace buffer check → Overtime token issuance → Statutory muster roll sync.
4. **Absence Governance:** Leave request submission → Tier-1 Manager endorsement → Tier-2 HR sign-off → Quota meter balance deduction.
5. **Statutory Payroll & Disbursal:** Attendance + overtime ingestion → Gross pay → FBR 2025–27 tax bracket math → EOBI/PF deductions → 6 Bank CSV batch generation → Payslip PDF delivery.
6. **Lifecycle Exit & Settlements:** Resignation / termination → 4-Gate departmental clearance → Statutory 30/26 gratuity voucher → Final & Full (F&F) sign-off.

2. Technology Stack & Infrastructure

Layer	Technologies & Libraries	Architectural Purpose & Implementation Details
Frontend Core	Vanilla HTML5, ES6+ JavaScript, CSS3 Design Tokens	Zero-build modern SPA. No heavy framework overhead; delivers sub-100ms client route transitions and extreme memory efficiency.
Styling & Design	Custom CSS Properties, Glassmorphism, CSS Grid/Flexbox	Dual-Theme Engine (Obsidian Dark Mode & Crisp Light Mode) persisted in localStorage; high-density compact table switcher.
Backend Core	Node.js (v20+), Express.js (v4.22.3)	RESTful API layer serving 72 endpoints, static asset middleware, and SPA HTML5 fallback routing.
Database & ORM	Prisma ORM (v6.4.1), PostgreSQL, Local JSON Store	Dual storage: Prisma schema supporting 103 models on PostgreSQL (Neon / Supabase / Vercel Postgres) + ACID-compliant file JSON store for offline/standalone execution.
Real-Time Engine	WebSocket (`ws` v8.22.0), Server-Sent Events (SSE)	Bidirectional live channel messaging, online presence indicators, and real-time database change broadcasts.
Biometric Gateway	`node-zklib` (v1.3.0), Raw TCP/IP Socket Adapter	Native hardware communication with physical ZKTeco, SilkID, and IP biometric attendance devices on port 4370.
AI Intelligence	Google Gemini 3.5 Flash-lite, Google AI Studio SDK	Rapid ~1.3s AI HR co-pilot, automated policy Q&A, resume ATS scoring, and local fallback knowledge base.
Security & Auth	JWT (`jsonwebtoken` v9.0.2), `bcryptjs` , `helmet`	Stateless authentication tokens, HTTP security headers, clickjacking defense, brute-force throttling, and idle session auto-locks.
Email Disbursal	Nodemailer (v10.0.10)	Automated email delivery of digital payslips, leave decisions, warning letters, and daily attendance audit digests.
PWA & Mobile	Service Worker (`sw.js`), Web App Manifest	Progressive Web App installability on iOS/Android/Windows; offline asset caching and cryptographic punch queues.

3. Project Structure & Architecture Pattern

3.1 Architecture Pattern

HRM Pro employs an **Offline-First Single Page Application (SPA) Architecture** paired with a **Modular Monolith REST API backend**:

- **Client Side:** Event-driven Vanilla JS modules mounted to a central `#app` container via `App.navigate(module, subView)` . Dynamic UI rendering with strict XSS sanitization (`Utils.escapeHtml`).
- **Server Side:** Controller-route pattern organized under `server/src/routes/` , backed by transactional services in `server/src/services/` and Prisma client.
- **Dual Static Synchronization:** The root directory (`js/` , `css/` , `index.html`) is strictly synchronized with `public/` for production/Vercel static build deployments.

3.2 Directory Layout

Path / Folder	Key Files	Functional Responsibility
`/` (Root)	`index.html` , `package.json` , `sw.js`	Application shell, HTML structure, vendor CDN inclusions, service worker registration.
`/js`	`app.js` , `auth.js` , `data.js` , `dashboard.js` , `employees.js` , `payroll.js` , `chat.js`	41 client JavaScript modules housing all application logic, UI renderers, state management, and API integrations.
`/css`	`main.css` , `landing-automark.css` , `landing-theme.css`	Design system tokens, typography, Obsidian dark and light themes, responsive media queries.
`/server/src`	`server.js` , `db.js` , `websocket.js`	Express HTTP entrypoint, WebSocket gateway, server-side security headers.

Path / Folder	Key Files	Functional Responsibility
`/server/src/routes`	`auth.js`, `employees.js`, `payroll.js`, `leaves.js`, `sync.js`, `chat.js`	72 REST endpoints handling authentication, workforce CRUD, tax calculations, and database sync.
`/server/src/services`	`store.js`, `taxEngine.js`, `emailService.js`	Persistent server file store, FBR progressive tax computation engine, SMTP mailer.
`/server/src/jobs`	`dailyAttendanceSummary.js`	Automated cron job generating daily managerial attendance digests.
`/server/prisma`	`schema.prisma`, `seed.js`	103 Prisma models, PostgreSQL relations, database seeding scripts.
`/scripts`	`ci-audit.js`, `system-wide-health-audit.js`, `generate-pdf.js`	Automated testing suites, CI/CD quality assurance gates, documentation generators.
`/public`	`public/js/`, `public/css/`, `public/index.html`	Exact mirror of root assets used for Vercel and production cloud builds.

4. User Roles & RBAC Permissions Matrix

4.1 Role Hierarchy & Scope Definitions

The system enforces least-privilege role-based access control (RBAC) across 7 distinct personas with multi-tier scoping:

- `ALL` — Unrestricted visibility across all corporate subsidiaries and employee master files.
- `TEAM` — Scoped strictly to direct and recursive indirect subordinates in the organizational hierarchy.
- `SELF` — Scoped strictly to the authenticated user's own profile, attendance, leaves, and payslips.

Role Identifier	Display Title	Data Scope	Access Level & Key Permissions
`superadmin`	Super Administrator	`ALL` (Holding-wide)	Unrestricted universal authority. Manages legal entities, NTNs, global settings, audit logs, and system RBAC matrix.
`hr_manager`	HR Director / Manager	`ALL` (Entity-scoped)	Workforce lifecycle master, document verification, final leave authorization, salary generation, and exit settlements.
`dept_manager`	Department Head	`TEAM` (Subordinates)	Department attendance monitoring, Tier-1 leave endorsement, shift swap approvals, interview evaluations, and team OKRs.
`payroll_accountant`	Payroll & Tax Accountant	`ALL` (Finance only)	Salary calculation, FBR tax slab adjustments, 1Link/PayPak bank batches, expense claim reimbursements. Cannot edit employee contracts.
`employee`	Regular Employee	`SELF` (Own Profile)	Self-service web punch clock, leave applications, expense submissions, confidential helpdesk tickets, personal payslip downloads.
`onboarding`	Onboarding Recruit	`SELF` (Induction)	Restricted induction portal. Document uploading, emergency contacts submission, company policy acknowledgments.
`junior_hr`	Junior HR Officer	`ALL` (Read-Only)	Can view employee directories and log attendance; approval actions, salary generation, and quota edits are revoked.

5. Complete Feature Inventory (19 Modules)

Module Name	Core Purpose	User Flow & Views	Tables / Models	Key APIs & Methods
1. Executive Dashboard	Executive command center with live punch clock, KPI sparklines, priority inbox, headlines.	`#dashboard` → View telemetry → Action priority items → Multi-company filter.	`attendance`, `employees`, `companies`, `payroll`	`Dashboard.render()`, `/api/attendance/today-summary`
2. Employee Lifecycle & e-DMS	4-stage workforce lifecycle: Directory Roster, e-DMS Vault, Letters Hub, Life Events.	`#employees` → Directory → Org Chart → e-DMS Vault → Document Expiry Radar.	`employees`, `documents`, `dependents`, `life_events`	`Employees.render()`, `/api/employees`
3. Biometric Attendance Gateway	Hardware terminal sync, machine simulator, offline PWA queue, statutory muster roll, geofence.	`#attendance` → Daily logs → Terminal simulator → Muster roll → Geofence radar.	`attendance`, `shifts`, `biometric_devices`, `shift_roster`	`Attendance.render()`, `/api/attendance/punch`
4. Leave Approvals & Balances	Statutory leave quotas, multi-tier approvals, gazetted holidays, encashment math.	`#leaves` → Request Leave → Manager Endorsement → HR Sign-Off → Ledger update.	`leave_requests`, `leave_types`, `leave_balances`, `holidays`	`Leaves.render()`, `/api/leaves/requests`
5. Statutory Payroll & FBR Taxes	Pakistan FBR 2025–27 tax slabs, EOBI/PF ledgers, 6 bank disbursal batch formats, payslip PDFs.	`#payroll` → Process Salary → Audit Slabs → Export 1Link/PayPak → Send Slips.	`payroll`, `tax_table`, `eobi_ledger`, `salary_slip_items`	`Payroll.render()`, `/api/payroll/generate`

Module Name	Core Purpose	User Flow & Views	Tables / Models	Key APIs & Methods
6. Recruitment ATS & Onboarding	5-stage applicant tracking Kanban, 10-criteria rubric scoring, digital offer letters.	`#recruitment` → Job Postings → Applicant Kanban → Scorecard → Hired to Employee.	`job_requisitions`, `candidates`, `interview_scorecards`, `offer_letters`	`Recruitment.render()`, `/api/jobs`
7. Performance & 9-Box Matrix	SMART OKRs, 360° reviews, 9-box talent matrix for leadership succession, LMS competencies.	`#performance` → Goals & KPIs → 360 Reviews → 9-Box Grid → Succession.	`performance_goals`, `performance_reviews`, `feedback_360`, `appraisals`	`Performance.render()`
8. Learning Management (LMS)	Course catalog, training calendars, managerial nominations, QR verification, certificate safe.	`#performance` → LMS Tab → Course Catalog → Enroll → Certificate Upload.	`courses_lms`, `course_enrollments`, `training_sessions`, `certificates`	`Performance.renderLmsTab()`
9. Team Chatbox & Collab Hub	Microsoft Teams-style collaboration: department channels, direct messages, meeting room launcher.	`#chat` → Select Channel → Send Message → Pin Document → Launch Meeting.	`chat_channels`, `chat_messages`, `chat_meetings`	`Chat.render()`, `/api/chat/messages`, WebSocket
10. Asset Management	Hardware inventory, laptop assignments, return condition tracking, warranty audits.	`#assets` → Add Asset → Assign Employee → Maintenance Log → Return Check.	`assets`, `asset_categories`, `asset_assignments`	`Assets.render()`
11. Expense Claims & Travel	Reimbursement lodging, multi-receipt attachments, 2-tier managerial and finance sign-offs.	`#expenses` → New Claim → Upload Receipt → Manager Approval → Finance Payout.	`expense_claims`, `travel_requests`, `expense_categories`	`Expenses.render()`
12. Helpdesk & Grievance Hub	IT support tickets, HR inquiries, priority SLAs, confidential whistleblower grievances.	`#helpdesk` → Lodge Ticket → Assign Agent → Status Updates → Resolution.	`helpdesk_tickets`, `notifications`	`Helpdesk.render()`
13. Company Events & Townhalls	Gazetted holiday calendar sync, corporate events, announcements, birthday alerts.	`#events` → View Calendar → Create Event → Broadcast Notification.	`company_events`, `holidays`, `announcements`	`Events.render()`
14. Reports & Executive BI	Workforce demographics, turnover rates, FBR tax schedules, bank audit checksums, CSV exports.	`#reports` → Select Report → Apply Date Range → Export CSV/Print PDF.	Consolidated aggregation across all tables	`Reports.render()`
15. Multi-Company Holdings	Model A corporate holdings, subsidiary legal scoping, multi-NTN profiles, entity switcher.	`#companies` → Company Roster → Scoping Configuration → Global Switcher.	`companies`, `organizations`, `business_units`	`Company.render()`, `/api/companies`
16. Exit & Gratuity Settlements	Pakistan statutory 30/26 gratuity formula, 4-gate clearance workflows, audit-ready F&F vouchers.	`#settlement` → Initiate Exit → 4-Gate Sign-Off → Calculate Gratuity → Print Voucher.	`settlements`, `exit_clearances`, `gratuity_pool`	`Settlement.render()`, `/api/settlements`
17. Master Data Administration	Departments, designations, branch geofences, shifts, corporate hierarchy, immutable audit logs.	`#administration` → Configure Departments → Manage Shifts → Audit Logs.	`departments`, `designations`, `branches`, `audit_logs`	`Administration.render()`, `/api/admin`

Module Name	Core Purpose	User Flow & Views	Tables / Models	Key APIs & Methods
18. System Settings & RBAC	Permissions matrix editor, theme customization, database export/import, security policies.	`#settings` → General Settings → Roles & Permissions Matrix → Backup DB.	`system_settings`, `roles`, `permissions`, `role_permissions`	`Settings.render()`
19. Google Gemini AI Co-Pilot	Sub-1.5s AI HR assistant, labor law policy Q&A, resume ATS parsing, local fallback knowledge base.	Click floating Co-Pilot → Ask policy question → Immediate contextual guidance.	`hrm_gemini_api_key`, Local Policy KB	`GeminiService.ask()`, `HRAssistant.render()`

6. Database Architecture & Data Relationships

6.1 Database Schema & Entity Relationships

The core schema revolves around the central `Employee` entity with relational foreign keys linking workforce identity across operational domains:

- **Workforce Hierarchy:** `Department (1) → (N) Designation (1) → (N) Employee`. Self-referential `Employee.managerId → Employee.id` constructs the infinite-depth organizational tree.
- **Multi-Company Scoping:** `Company (1) → (N) Employee` and `Company (1) → (N) Branch` enforcing strict data partitioning.
- **Time & Attendance:** `Employee (1) → (N) Attendance`, `Shift (1) → (N) Employee`, and `BiometricDevice (1) → (N) AttendanceLog`.
- **Leave Ledgers:** `LeaveType (1) → (N) LeaveBalance` and `Employee (1) → (N) LeaveRequest`.
- **Compensation:** `Employee (1) → (N) Payroll → (N) SalarySlipItem`. Linked to `TaxTable` and `EobiLedger`.
- **Exit Clearance:** `Employee (1) → (1) Settlement → (4) ExitClearance` (HR, IT, Finance, Administration).
- **Security & RBAC:** `User (1) → (N) UserRole → (1) Role (1) → (N) RolePermission → (1) Permission`.

6.2 Key Data Fields & Structural Constraints

Table / Collection	Primary Key	Foreign Keys	Key Business Attributes
`employees`	`id` (Int / UUID)	`departmentId`, `designationId`, `companyId`, `managerId`, `shiftId`	`empNo` (Unique), `cnic`, `joiningDate`, `salary`, `status` (active/terminated), `bloodGroup`
`attendance`	`id` (Int / UUID)	`employeeId`, `companyId`, `deviceId`	`date`, `inTime`, `outTime`, `dutyHours`, `status` (Present/Late/HalfDay/Absent), `verifiedBy`
`leave_requests`	`id` (Int / UUID)	`employeeId`, `leaveTypeId`, `managerId`	`startDate`, `endDate`, `days`, `status` (Pending/Approved/Rejected), `gate1Status`, `gate2Status`
`payroll`	`id` (Int / UUID)	`employeeId`, `companyId`	`month`, `year`, `grossSalary`, `incomeTax`, `eobiDeduction`, `pfDeduction`, `netSalary`, `status`
`settlements`	`id` (Int / UUID)	`employeeId`, `companyId`	`exitDate`, `gratuityAmount`, `leaveEncashment`, `deductions`, `netPayable`, `clearanceStatus`
`audit_logs`	`id` (Int / UUID)	`userId`	`timestamp`, `action` (LOGIN/CREATE/UPDATE/DELETE), `module`, `details`, `ipAddress`

7. Frontend Architecture, UI Components & State Management

7.1 Component Organization & Navigation

- **Global Router (`App.navigate`):** Client hash-based router (`#dashboard`, `#employees`, `#payroll`, etc.) managing browser history without page reloads.
- **Two-Tier Persistent Navigation Dock:** Top tier houses primary functional modules; bottom sub-navigation tier exposes contextual sub-views (e.g. `Directory`, `Org Chart`, `e-DMS`).
- **Spotlight Command Palette (`Ctrl+K`):** Instant omnibar search across all employees, pages, actions, and settings.
- **Modal & Toast Notification Engine:** Centralized `Modal.show({ title, body, footer })` and toast notification manager (`Toast.show(msg, type)`).

7.2 Client State Management

Client state is managed by the reactive **`DB` Store Object** in `js/data.js`:

- Data mutations occur via `DB.insert(table, row)`, `DB.update(table, id, data)`, and `DB.delete(table, id)`.
- Mutations automatically update local memory cache, persist to `localStorage`, write to the audit trail (`DB.log`), and queue background synchronization to the Express server.
- Cross-tab synchronization is maintained via browser `storage` events, keeping open tabs synchronized across personas.

8. Backend Analysis & Server Services

8.1 RESTful API Endpoints & Routing Architecture

The Express.js application mounts 72 structured REST endpoints organized across 13 dedicated routers:

Router Module	Base Path	Endpoints & Operations
`auth.js`	`/api/auth`	`POST /login`, `GET /me`, `POST /change-password`
`employees.js`	`/api/employees`	`GET /`, `GET /:id`, `POST /`, `PUT /:id`, `DELETE /:id`
`attendance.js`	`/api/attendance`	`GET /`, `GET /today-summary`, `POST /punch`, `GET /biometric-sync`, `POST /biometric-sync`
`leaves.js`	`/api/leaves`	`GET /balances`, `GET /requests`, `POST /requests`, `PUT /requests/:id/status`
`payroll.js`	`/api/payroll`	`GET /`, `GET /tax-slabs`, `POST /tax-slabs`, `POST /generate`, `PUT /:id/status`
`settlements.js`	`/api/settlements`	`GET /`, `POST /`, `PUT /:id`, `DELETE /:id`, `POST /:id/clearance`
`companies.js`	`/api/companies`	`GET /`, `POST /`, `PUT /:id`, `DELETE /:id`, `POST /transfer`
`chat.js`	`/api/chat`	`GET /channels`, `POST /messages`, `POST /upload`, `GET /meetings`, `POST /meetings`
`email.js`	`/api/email`	`POST /payslip`, `POST /leave-request`, `POST /leave-decision`, `POST /welcome`
`sync.js`	`/api/sync`	`GET /all`, `POST /set`, `POST /batch`, `POST /apply`, `GET /stream` (SSE)
`jobs.js`	`/api/jobs`	`GET /daily-attendance-summary/status`, `POST /daily-attendance-summary/run`, `GET /cron/...`

8.2 Background Jobs & Workers

- **Daily Attendance Digest Worker (`jobs/dailyAttendanceSummary.js`):** Ticker checking shifts and scheduled hours; automatically compiles daily absent/late tallies and dispatches email digests to department managers and HR directors.
- **Document Expiry Radar:** Scheduled scan verifying contract end dates, CNIC validity, and professional certifications with 30/60/90-day alert triggers.

9. External Integrations

- **Google Gemini 3.5 AI Studio API:** Generative AI inference for natural language labor inquiries and automated resume ATS candidate scoring.
- **ZKTeco & SilkID Biometric Fleets:** Raw TCP/IP communication over port 4370 for automated biometric clock-in ingestion.
- **Commercial Bank Disbursal Gateways:** Generates formatted payment batch CSV/Excel files for 1Link, PayPak, Habib Bank Limited (HBL), Meezan Bank, and Bank Alfalah.
- **Supabase Cloud PostgreSQL:** Managed cloud database host with SSL connection pooling.
- **SMTP Relays (Nodemailer):** Email dispatches via configurable corporate SMTP servers.
- **PWA Service Worker:** Background asset caching and offline execution support.

10. Complete Application User Journey

1. **Portal Authentication:** User accesses `#login` → Selects HRM Suite or Team Chatbox → Enters credentials or clicks 1-tap demo persona → Session verified → Lands on `#dashboard`.
2. **Executive Dashboard:** Review headlines, live punch clock widget, pending action items in Priority Inbox → Check workforce telemetry.
3. **Record Creation:** Navigate to Employees/Recruitment/Leaves → Open modal form → Validation checks → Submit → Written to DB → Audit logged → Synced via WebSocket.
4. **Two-Tier Approvals:** Line manager endorses Tier-1 → HR Director grants Tier-2 final sign-off → Auto-notification dispatched.
5. **Payroll Run:** Finance opens `#payroll` → Ingests attendance → Computes FBR tax slabs → Disburses 1Link batch → Delivers payslips.
6. **Safe Logout:** User clicks avatar menu → Logout → Session purged from memory and localStorage → Redirects to Login screen.

11. Known Issues & Architectural Risk Areas

Risk Domain	Identified Architectural Challenge	Recommended Mitigation Strategy
Dual Directory Sync	Modifying root `js/` or `css/` without mirroring to `public/` creates drift on Vercel production static builds.	Enforce automated pre-commit hook or CI check verifying byte-for-byte synchronization between root and public directories.
Prisma DB Fallback	If remote PostgreSQL connection experiences latency or downtime, backend relies on local JSON store.	Configure Redis caching layer and circuit breakers on database pools for high-availability cloud deployments.
Client-Side Store Size	Extensive audit logs and attendance histories stored in browser `localStorage` can exceed the 5MB browser quota.	Implement IndexedDB storage driver for large document vaults and archive older attendance logs to the backend database.
Gemini API Rate Limits	High-frequency queries to Google Gemini endpoints could exhaust quota during peak business hours.	Model routing already falls back gracefully across 7 model priority tiers and local knowledge bases; implement token rate caching.

12. Comprehensive QA Testing Map & Verification Strategy

Test Suite / Module	Critical User Flows	High-Risk Features & Edge Cases	Verification Command
CI/CD Quality Gate	15 Core application routes navigation and DOM integrity verification.	Zero undefined template strings, zero NaN arithmetic values, zero uncaught JS exceptions, zero layout overflows.	`node scripts/ci-audit.js`
System Feature Audit	All 11 navbar modules, sub-views, and database schemas.	Event handler binding, table schemas, route mounting, session purging, responsive layout at 100%/125%/150% zoom.	`node scripts/system-wide-health-audit.js`
FBR Payroll Engine	End-of-month salary run across all 8 progressive tax brackets (0% up to 35%).	Mid-year increment adjustments, custom Provident Fund % rates, statutory EOBI thresholds, 6 bank CSV checksums.	`node scripts/staging-payroll-fbr-run.js`
Statutory Gratuity	Full & Final (F&F) employee offboarding settlement vouchers.	30/26 Gratuity formula: `(Last Gross ÷ 26 × 30) × Years`, multi-gate clearances across HR, IT, Finance, Admin.	`node scripts/verify-settlement-engine.js`
Multi-Company Scoping	Model A corporate holdings data isolation.	Subsidiary HR managers cannot view sister company employees; Group Super Admins maintain unified cross-NTN oversight.	`node scripts/test-all-11-navbar-modules.js`
Landing Page DOM	Marketing SaaS landing page, tax calculator, 10 pillars showcase.	Tax calculations match 2026–27 brackets, dual theme switcher functions, zero third-party brand leaks.	`node scripts/verify-landing-dom.js`